Monodip Bhattacharya

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Education

Dr. BC Roy Engineering College, Durgapur — Durgapur, India
B.Tech in Electronics and Communication Engineering — *Aug 2022 – July 2026*
- CGPA: 6.63/10

Indira Gandhi Memorial High School — Kolkata, India
Class XII – CBSE — *March 2022*

St. Stephen's School — Kolkata, India
Class X – ICSE — *March 2020*

Technical Skills

Languages: JavaScript (ES6+), Python, TypeScript, HTML5, CSS3, SQL
Frameworks & Libraries: React.js, Node.js, Express.js, TailwindCSS, Zustand
Databases: MongoDB, MySQL
Tools & Platforms: Git, GitHub, REST APIs, JWT, bcrypt, Arduino, IoT Protocols
Concepts: Data Structures, Full-Stack Development, Real-Time Systems, Firmware Development

Internships

Embedded Systems & IoT Intern — June 2025 – July 2025
IC Design and Fabrication Center, Dept. of ECE, Jadavpur University — *Kolkata, India*
- Programmed Arduino microcontrollers to interface with sensors and transmit data over IoT protocols, building end-to-end hardware-software pipelines covering circuit design, firmware development, and IoT data flow architecture.
- Delivered a complete real-time IoT project within a 2-month timeline; received official certification upon project completion.

Projects

Donor | *React.js, Node.js, Express, MongoDB, Socket.io, TailwindCSS, Zustand* | GitHub | Live — 2026
- Built a real-time blood donation coordination platform that automatically matches compatible donors within a 15km radius of hospital emergencies, using MongoDB geospatial indexing for fast location-based queries.
- Engineered real-time WebSocket notifications with Socket.io to instantly alert matched donors, and built an in-app chat system for live donor-requester coordination, secured with JWT authentication in HTTP-only cookies and bcrypt password hashing.
- Delivered a fully responsive, mobile-first UI with React, Framer Motion, and TailwindCSS v4, ensuring a smooth experience across phones, tablets, and desktop.

Chatify | *React.js, Node.js, Express, MongoDB, Socket.io, TailwindCSS, Cloudinary, Zustand* | GitHub | Live — 2026
- Built a production-ready real-time chat application with Socket.io and Node.js, handling bidirectional data synchronization and live messaging across concurrent users.
- Engineered a scalable MongoDB/Mongoose database layer with cursor-based pagination and infinite scrolling to optimize query performance and reduce payload sizes.
- Secured auth with JWT in HTTP-only cookies (XSS prevention); built read receipts, typing indicators, and message editing; integrated Cloudinary for media hosting and a responsive UI with 30+ themes via TailwindCSS and DaisyUI.

AI Resume Analyzer | *React, Vite, PDF.js, Puter.js AI, REST APIs* | GitHub | Live — 2025
- Built an AI-powered resume analysis tool with React, Vite, and PDF.js that parses uploaded PDFs and delivers ATS scores, grammar feedback, keyword suggestions, and recommended job roles via an analytics dashboard powered by Puter.js AI.

Additional

Interests: Football, Music, Reading, Travelling
Soft Skills: Problem-solving, Analytical Thinking, Time Management, Team Collaboration, Communication